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B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)

EXAMINATION JUNE 2023

Compiler Design

[Time:3:00 Hours]

[Max. Marks:100]

- Instructions:**
1. Make suitable assumptions if required. State your assumptions.
 2. Attempt any two Questions from PART-A and PART-B and one question from PART-C.

Part A

- Q1 a) Explain the different phases of compiler construction and the components involved (10 marks) with a diagram and an example.
- b) What are lexical errors and syntactic errors? Explain the error recovery strategies (10 marks) for both types of errors.
- Q2 a) Compute the FIRST and FOLLOW sets of the following grammar. (5 marks)
- S → aAbB | aAaB
A → AS | a | b
B → BSa | a | c
- b) Derive the DFA implementation of the Lexical Analyser for the following LEX (10 marks)
- ```
%{
 enum = {AB, B, BCC};
}%
%%
b {return (B); }
b+c+ {return (BCC); }
a*b+ {return (AB); }
%%
```
- c) Discuss compiler writing tools. (5 marks)
- Q3 a) Explain error recovery in YACC. Write a YACC specification for a Desk Calculator with error recovery. (10 marks)
- b) Explain the process of bootstrapping and porting with the help of T-diagrams. (5 marks)
- c) Discuss the organizational issues of a two pass assembler with a neat diagram. (5 marks)

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**Part B**

- Q4 a) Explain the issues in the design of a code generator. (6 marks)
- b) Construct the operator precedence parsing table for the grammar (8 marks)  
 $S \rightarrow (L) \mid a$   
 $L \rightarrow L, S \mid S$
- c) Convert the following statement into quadruple, triple and indirect triple representation. (6 marks)  
 $\text{Position} = \text{initial} + \text{rate} * A[i]$
- Q5 a) Explain the principal sources of optimization? (10 marks)
- b) What are the applications of dags? Construct a dag for: (10 marks)  
 $t1 = a + b$   
 $x = t1$   
 $t2 = a - b$   
 $y = t2$   
 $z = x + y$
- Q6 a) List and explain the data structures used to implement a symbol table. (5 marks)
- b) Why is it necessary to break the intermediate code into basic blocks? Translate the following code fragment to 3-address statements and draw the flow-graph. (10 marks)  
 $\text{count} = 0;$   
 $\text{for}(\text{int } i=0; i < 10; i++) \{$   
 $\quad \text{count} += 1;$   
 $\quad \text{max} = 100$   
 $\quad x[i] = \text{count};$   
 $\}$
- c) Write the syntax-directed translation scheme for the switch statement. (5 marks)

**Part C**

- Q7 a) Explain applications of compiler technology. (8 marks)
- b) Write the regular expression and draw the transition diagram for (4 marks)  
 i) Identifier in C  
 ii) White space in C
- c) Explain top-down parsing. (8 marks)

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- Q8 a) Define and explain token, pattern, and lexeme. **(6 marks)**
- b) Write a short note on the LEX tool. **(5 marks)**
- c) Write the translation scheme to produce three-address code for assignments. **(5 marks)**
- d) What was the impact on Compilers writers with the evolution of programming languages? **(4 marks)**